

Accuracy of the maxillomandibular relationship at centric relation position recorded by using 3 different intraoral scanners with or without an optical jaw tracking system: An *in vivo* pilot study

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ABSTRACT

Purpose: To measure the accuracy (trueness and precision) of the maxillomandibular relationship at centric relation position recorded by using 3 different intraoral scanners with or without an optical jaw tracking system.

Material and methods: A completely dentate volunteer was selected. Seven groups were generated: conventional procedure (control group), 3 IOSs: Trios4 (Trios4 group), Itero Element 5D Plus (Itero group), i700 (i700 group), and 3 groups with a jaw tracking system for each corresponding IOS system (Modjaw-Trios4, Modjaw-iTero, and Modjaw-i700 groups) ($n = 10$). In the control group, casts were mounted on an articulator (Panadent) using a face bow and a CR record captured with the Kois deprogrammer (KD). The casts were digitized by using a scanner (T710) (control files). In the Trios4 group, intraoral scans were obtained by using the corresponding IOS and duplicated 10 times. The KD was used to obtain a bilateral occlusal record at CR position. These same procedures were followed for the Itero and i700 groups. In the Modjaw-Trios4 group, the intraoral scans acquired by using the corresponding IOS at MIP were imported into the jaw tracking program. The KD was used to record the CR relationship. For acquiring the specimens in the Modjaw-Itero and Modjaw-i700 groups, the same procedures were followed as in the Modjaw-Trios4 group, with the scans obtained with the Itero and i700 scanners respectively. The articulated virtual casts of each group were exported. Thirty-six inter-landmark linear measurements were used to calculate the discrepancies between the control and experimental scans. The data were analyzed by using 2-way ANOVA followed the pairwise comparison Tukey tests ($\alpha = 0.05$).

Results: Significant trueness and precision discrepancies were found among the groups tested ($P < .001$). The Modjaw-i700, Modjaw-iTero, Modjaw-Trios4, and i700 groups obtained the best trueness and precision among the groups tested, and the iTero and Trios4 groups obtained the worst trueness. The iTero group obtained the worst precision among the groups tested ($P > .05$).

Conclusions: The maxillomandibular relationship recorded was influenced by the technique selected. Except for the i700 IOS system, the optical jaw tracking system tested improved the trueness value of the maxillomandibular relationship recorded at CR position when compared with the corresponding IOS.

1. Introduction

Conventionally, diagnostic stone casts mounted on a physical articulator at centric relation (CR) [1] position might be required in different

diagnostic and prosthodontic treatment procedures [2]. Many factors have been identified that can impact the accuracy and reproducibility of the CR recording techniques including body position, time of the day, sleep, psychological variables that can modify muscle tonus, stress,

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proprioception, occlusal changes, muscle spasms, and temporomandibular joint dysfunction [3–7].

Methods used to record the maxillomandibular relationship in CR position include graphical methods, functional recordings, cephalometric techniques, direct interocclusal records, Lucia jig, leaf gage technique, and the Kois deprogrammer (KD) method [8–12]. Recording digital methods may combine intraoral scanners (IOSs) and optical jaw tracking systems [13,14].

Intraoral scanners (IOSs) provide a digital alternative to obtain virtual diagnostic and definitive casts, as well as acquiring virtual occlusal records to record the maxillomandibular relationship [15–20]. However, the accuracy of the maxillomandibular relationship acquired at CR position by using IOSs systems remains uncertain [17–20]. Knowing the factors that can decrease scanning accuracy of IOSs is fundamental for maximizing the accuracy of the virtual casts and the maxillomandibular relationship. Factors related to operator skills and decisions include calibration [21], scanning technology [22,23], ambient lighting conditions [24,25], scanning pattern [26,27], extension of the intraoral digital scan [28,29], and operator experience [30]. Factors related to the intraoral conditions of the patient, including intraoral humid conditions [31] and characteristics of the surface being digitized, also affect the accuracy of the virtual casts and records [32,33].

Dental literature has described different experimental or commercially available optical jaw tracking systems for recording both the static maxillomandibular relationship and the dynamic motion of the mandible [34–36]. However, the accuracy of these systems remains unclear. These systems require the use of virtual diagnostic casts in maximum intercuspation (MIP) obtained by using an IOS. Yet the influence of the IOS selection on the accuracy of the optical jaw tracking systems remains unknown.

The objective of the present pilot clinical study was to evaluate the accuracy of the maxillomandibular relationship captured at CR position obtained by using a conventional procedure (control group) and 3 IOSs (Trios4, 3Shape A/S; Itero Element 5D Plus, Align Technology; and i700, Medit) with or without an optical jaw tracking system (Modjaw; Modjaw). This pilot study aimed to assess the accuracy of the mandible position established with a Kois deprogrammer (KD), not the accuracy of the CR position per se. The null hypotheses were that there would be no maxillomandibular relationship discrepancies acquired at CR position obtained by using conventional or any of the digital methods tested.

2. Material and methods

2.1. Data acquisition

The research protocol scheme is presented in Fig. 1. The protocol was approved by a Institutional Review Board (IRB) committee (IRB: 2302899). A 41-year-old male volunteer was selected as the study participant. The participant had a good oral health with a complete second molar to second molar dentition without indirect restorations. The participant had no temporomandibular discomfort, sounds, or popping. The participant was able to comfortably chew gum or hard foods and have a mandibular range of motion within the normal limits (maximum opening of 54 mm).

Prior to data collection, an additively manufactured KD [12] and a 3D printed custom device for the jaw tracking system were fabricated [36]. The additively manufactured KD was delivered following a previously described technique [12]. Subsequently, the participant was positioned at 45° in a dental chair [6,7]. An anterior jig was created using a polyvinyl siloxane dental material (Virtual heavy and light

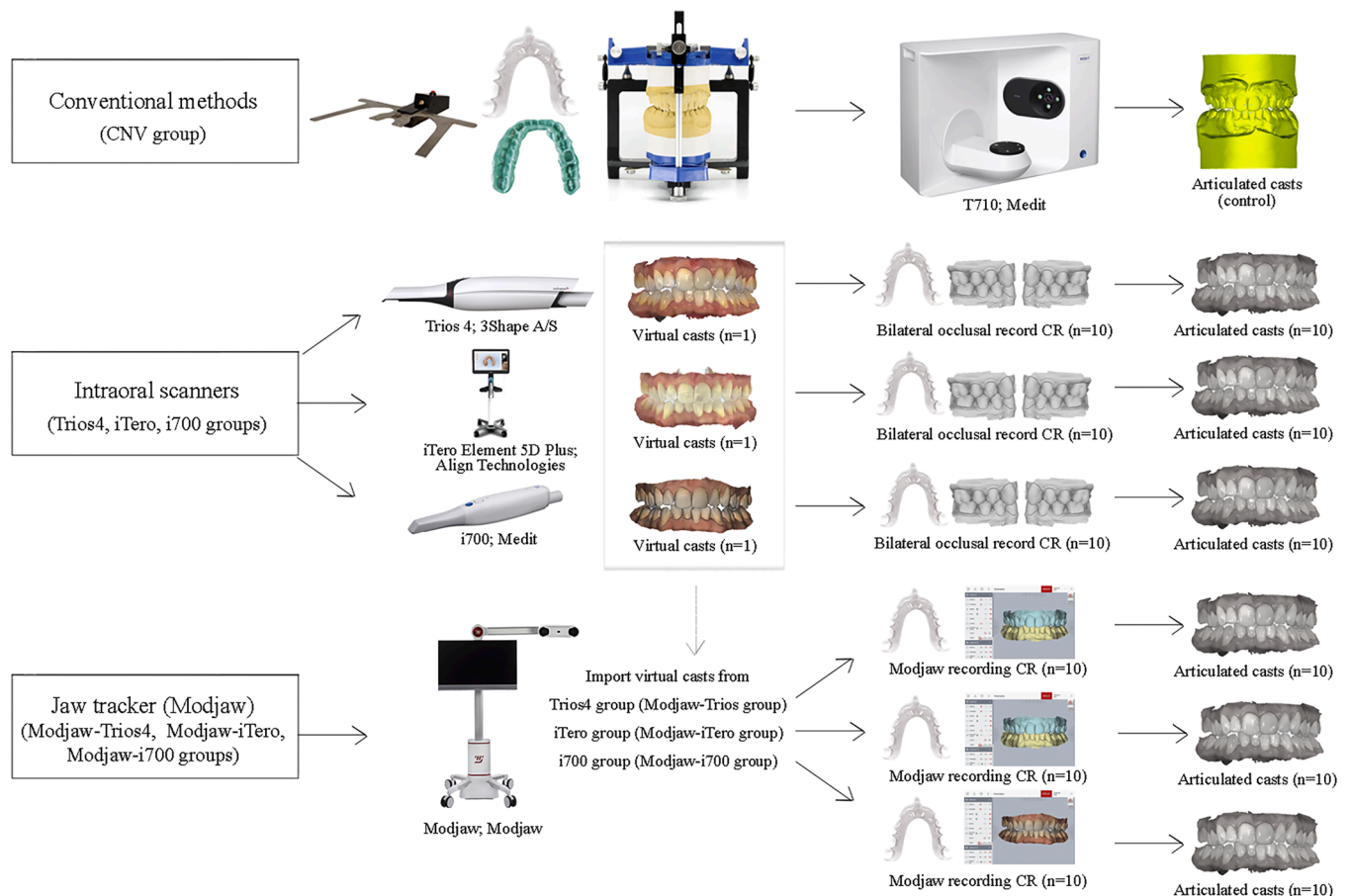


Fig. 1. Research methodology scheme for data acquisition procedures.

viscosity regular set; Ivoclar Vivadent). The purpose of this anterior jig was to facilitate the reproducibility of the mandible’s position when acquiring the occlusal records using the KD. Additionally, 3 printed custom devices for the optical jaw tracking system (Modjaw; Modjaw) were fabricated following a technique previously described [36].

Six groups were created based on the technique used to acquire the diagnostic casts and the maxillomandibular relationship at CR position: a conventional technique (CNV or control group); 3 IOSs: Trios4 (Trios4, wireless, v. 21.2.0; 3Shape A/S) (Trios group), Itero (Itero Element 5D Plus; Align Technology) (Itero group), and i700 (i700, wireless; Medit) (i700 group); and 3 groups using an optical jaw tracking system (Modjaw; Modjaw) (Modjaw group) which were based on the IOS used to obtain the scans imported into the jaw tracking system: Modjaw-Trios4, Modjaw-Itero, and Modjaw-i700 groups (*n* = 10) (Table 1). In the present pilot study, a sample size of 10 specimens was determined. All the data was obtained by a prosthodontist with 15 years of clinical experience and 10 years of experience with IOSs. The order of the groups was randomized by using a shuffled deck of cards. The records for only 1 group per day were acquired to avoid participant’s fatigue and all the data acquisition procedures were performed in the same range of morning hours.

In the CNV group, maxillary and mandibular polyvinyl siloxane (Virtual heavy and light viscosity regular set; Ivoclar Vivadent) impressions were obtained. The impressions were poured with Type IV dental stone (GC Fujirock EP; GC) after mixing 22 ml water with 100 g dental stone under vacuum for 30 s. The casts were recovered after the dental stone had completely set. A conventional face bow record (Kois Dento-Facial Analyzer; Panadent) was obtained and used to mount the maxillary stone cast on a semi-adjustable articulator (Panadent Articulator, PCH Magnetic Model; Panadent) with the 20° of condylar inclination and 1.5 mm immediate side shift (Preformed Condylar Axis Motion Analog, 1.5 mm; Panadent). The selected articulator had a fixed

intercondylar distance of 110 mm. The patient was positioned at 45° in the dental chair [6,7]. The KD and the polyvinyl siloxane anterior jig were positioned, and 3 polyvinyl siloxane (Futar & Futar Fast; Kettenbach Dental) occlusal records were acquired. One of the 3 records was used to mount the mandibular cast on the same semi-adjustable articulator, while the other 2 records were used to verify the mounting of the mandibular casts (Fig. 2). The mounted casts were then digitized by using a laboratory scanner (T710; Medit) following the manufacturer’s protocol. The reference standard tessellation language (STL) files were acquired.

In the Trios4 group, maxillary and mandibular intraoral digital scans were obtained by using an IOS (Trios 4, wireless, v.21.2.0; 3Shape A/S) under 1000-lux ambient lighting conditions (LX1330B Light Meter; Dr. Meter Digital Illuminance) [24,25]. The IOS was calibrated before acquiring the intraoral digital scans according to the manufacturer’s protocol [21]. The scanning protocol recommended by the manufacturer for acquiring intraoral digital scans in completely dentate patients was followed. The intraoral digital scans were duplicated 11 times. Ten of the duplicated intraoral digital scans were used to acquire the specimens for the Trios4 group and the last duplicated intraoral digital scan was imported into the jaw tracking system to obtain the specimens for the Modjaw-Trios4 group. The patient was positioned at 45° in the dental chair [6,7] and 10 consecutive bilateral occlusal records were obtained with the KD and the polyvinyl siloxane anterior jig positioned in the patient’s mouth. These records were obtained following the scanning pattern recommended by the manufacturer. Each occlusal record included 4 teeth, from the first molar to the canine, to maximize the accuracy of the virtual maxillomandibular relationship [20]. The IOS software program automatically aligned the maxillary and mandibular casts by using the bilateral occlusal records, without any manual intervention of the operator (Fig. 3A). The articulated maxillary and mandibular casts were exported in a STL file format.

For acquiring the specimens of the Itero and i700 groups, the same procedures were followed as in the Trios4 group but using the corresponding IOS of each group (Figs. 3BC). A total of 10 specimens in a STL file format per group were obtained.

For the Modjaw-Trios4 group, the mandibular tracker attached to one printed custom device [60] was positioned in the mandibular arch by using a bis-acryl interim dental material (Luxatemp Automix Plus; DMG). The optical jaw tracking system was calibrated following the manufacturer’s protocol. Subsequently, the maxillomandibular relationship was recorded at CR position with the KD and the polyvinyl siloxane anterior jig positioned in the patient’s mouth. The static

Table 1
Digitizing procedures performed on each group.

Group	Diagnostic casts	Occlusal record
CNV	Conventional stone casts obtained from conventional impression procedures using polyvinyl siloxane dental material (Virtual heavy and light viscosity regular set; Ivoclar Vivadent).	Polyvinyl siloxane (Futar & Futar Fast; Kettenbach Dental) bilateral occlusal records at CR position using the KD
Trios4	Maxillary and mandibular intraoral digital scan obtained by using an IOS (Trios 4, wireless, v. 21.2.0; 3Shape A/S)	Bilateral virtual occlusal record at CR position using the KD and an IOS (Trios 4, wireless, v. 21.2.0; 3Shape A/S)
iTero	Maxillary and mandibular intraoral digital scan obtained by using an IOS (iTero Element 5D Plus; Align Technology)	Bilateral virtual occlusal record at CR position with the KD and an IOS (iTero Element 5D Plus; Align Technology)
i700	Maxillary and mandibular intraoral digital scan obtained by using an IOS (i700 wireless; Medit)	Bilateral virtual occlusal record at CR position with the KD and an IOS (i700 wireless; Medit)
Modjaw-Trios4	Maxillary and mandibular intraoral digital scan obtained by using an IOS (Trios4; wireless, v. 21.2.0; 3Shape A/S)	Maxillomandibular relationship recorded at CR position with the KD and an optical jaw tracking system (Modjaw; Modjaw)
Modjaw-iTero	Maxillary and mandibular intraoral digital scan obtained by using an IOS (iTero Element 5D; Align Technology)	Maxillomandibular relationship recorded at CR position with the KD and an optical jaw tracking system (Modjaw; Modjaw)
Modjaw-i700	Maxillary and mandibular intraoral digital scan obtained by using an IOS (i700 wireless; Medit)	Maxillomandibular relationship recorded at CR position with the KD and an optical jaw tracking system (Modjaw; Modjaw)

CNV, conventional; KD, Kois deprogrammer; IOS, intraoral scanner.

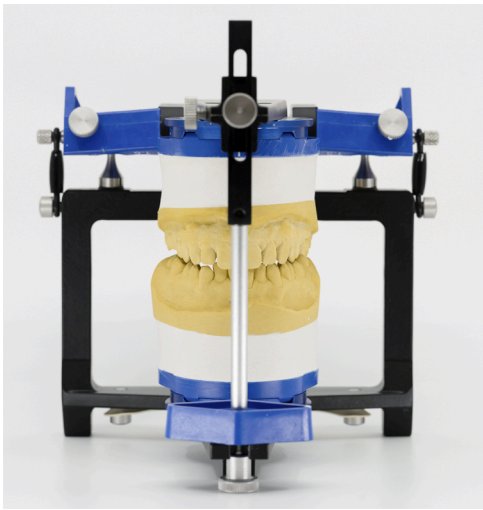


Fig. 2. Conventional procedures. Diagnostic stone casts mounted in CR position at the same vertical dimension captured using the KD. CR, centric relation; KD, Kois deprogrammer.

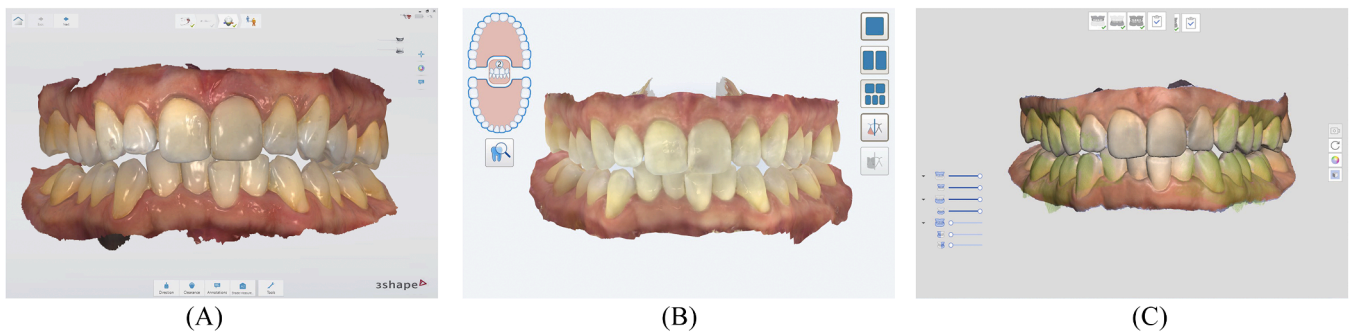


Fig. 3. Representative articulated virtual diagnostic casts at CR position obtained from intraoral digital scans and bilateral virtual occlusal records by using IOSs. A, Trios4 group. B, iTero group. C, i700 group. CR, centric relation; IOS, intraoral scanner.

relationship recorded with the optical jaw tracking system was exported in a STL file format (Fig. 4A). The procedure was repeated until 10 specimens were obtained.

For acquiring the specimens of the Modjaw-Itero and Modjaw-i700 groups, the same procedures were followed as in the Modjaw-Trios4 group but with the corresponding intraoral digital scans of each iTero and i700 groups imported into the jaw tracking system (Figs. 4BC). A total of 10 specimens in a STL file format per group were obtained.

2.2. Measurements

The STL files were imported into a reverse engineering software program (Geomagic Control X; 3D Systems). A total of 6 markers were digitally positioned on the maxillary and mandibular reference scans in the following positions: buccal surface of the right and left first molar, right and left canine, and right and left central incisors (Fig. 5A). A total of 36 inter-landmark linear measurements were made between the maxillary and mandibular reference casts (Fig. 5B). Then, an articulated maxillary and mandibular specimen was imported. In order to maintain the same location of the markers, the reference scans were aligned to the experimental scans without altering the position of the experimental scans; therefore, the maxillomandibular relationship of the experimental scans was unaltered (Fig. 5C). Once the experimental scans were aligned to the reference scans, the 12 features (6 maxillary and 6 mandibular) were copied from the reference mesh to the experimental scans [20]. Thus, the x-, y-, and z-coordinate of the markers were the same in the scans avoiding manual intervention of the operator and minimizing measurement errors [20]. This experimental model was used as master experimental model and was aligned to the remaining experimental scans. As the intraoral digital scans were the same in all the experimental groups, the virtual occlusal records were the variable element [20]. The marks of the reference scans were used as a reference to position the markers on the experimental scans. Subsequently, the

same inter-landmark linear measurements were performed. The same procedure was repeated until all the specimens of all the groups were measured.

For assessing the accuracy of the maxillomandibular relationship, inter-landmark linear measurements were performed on the reference and experimental scans as previously described. In this experiment trueness was defined as the average absolute discrepancy between inter-landmark measurements measured on reference and experimental scans [37,38], while precision was defined as the inter-landmark distance discrepancies between the measurements obtained in the reference and experimental scans [37,38].

2.3. Statistical analysis

The Kolmogorov-Smirnov test showed that the inter-landmark measurements data were normally distributed ($P > .05$). The data were analyzed by using 1-way ANOVA followed by pairwise comparison Tukey tests ($\alpha = 0.05$). All the data was analyzed using a statistical software program (SPSS Statistics, v26 for Windows; IBM Corp.).

3. Results

The maxillomandibular relationship discrepancies measured among the groups tested are presented in Table 2. Regarding trueness, 1-way ANOVA revealed significant trueness discrepancies among the groups tested ($df=5$, $MS=0.081286$, $F = 20.78$, $P < .001$) (Fig. 6A). Tukey pairwise comparison test showed significant trueness discrepancies among the groups tested (Fig. 6B). The Modjaw-i700, Modjaw-iTero, Modjaw-Trios4, and i700 groups obtained the best trueness mean (lowest maxillomandibular mean discrepancies) values among the groups tested and were not significantly different from each other ($P > .05$) (Table 2). The iTero and Trios4 groups obtained the worst trueness (highest maxillomandibular mean discrepancies) among the

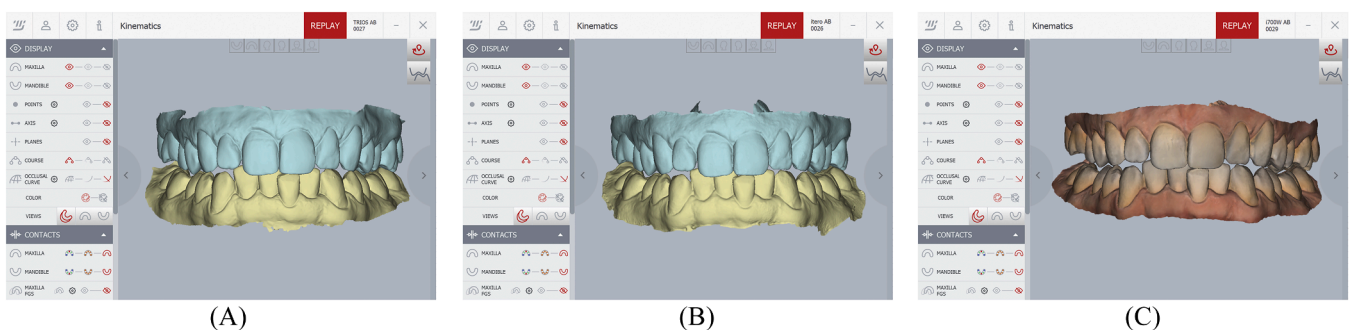


Fig. 4. Articulated virtual diagnostic casts at CR position obtained using an optical jaw tracking system (Modjaw; Modjaw) to import the acquired intraoral digital scans in MIP from different IOSs. A, Modjaw-Trios4 group. B, Modjaw-iTero group. C, Modjaw-i700 group. CR, centric relation; IOS, intraoral scanner; MIP, maximum intercuspation.

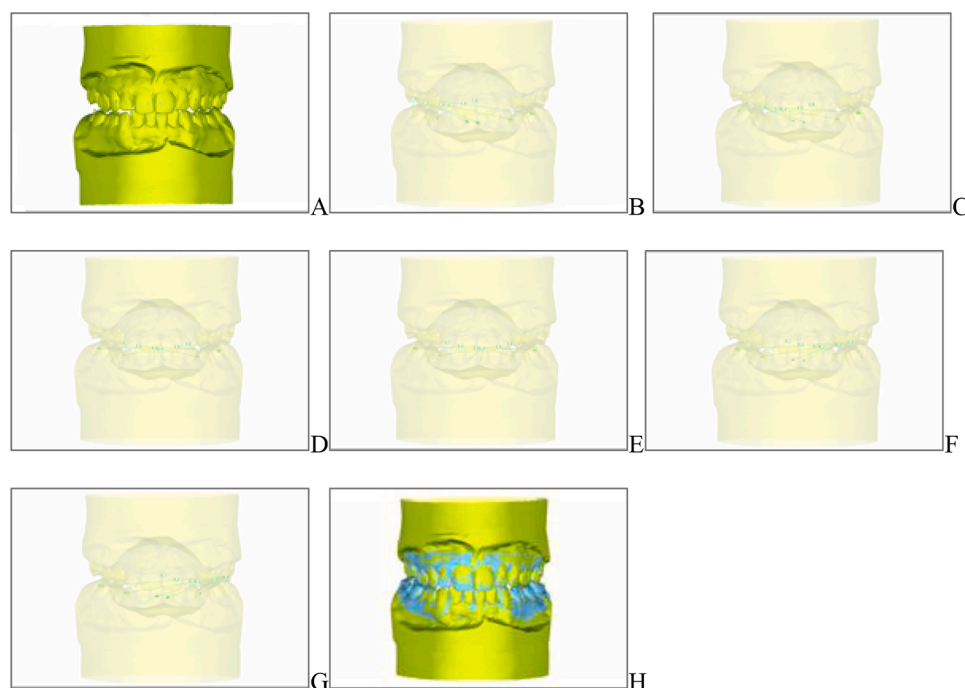


Fig. 5. Representative measurements completed. A, Six markers digitally positioned on each maxillary and mandibular reference scans. B, Representative inter-landmark measurements from the marker 1 of the maxillary arch to the 6 markers of the mandibular arch. C, Representative inter-landmark measurements from the marker 2 of the maxillary arch to the 6 markers of the mandibular arch. D, Representative inter-landmark measurements from the marker 3 of the maxillary arch to the 6 markers of the mandibular arch. E, Representative inter-landmark measurements from the marker 4 of the maxillary arch to the 6 markers of the mandibular arch. F, Representative inter-landmark measurements from the marker 5 of the maxillary arch to the 6 markers of the mandibular arch. G, Representative inter-landmark measurements from the marker 6 of the maxillary arch to the 6 markers of the mandibular arch. H, Experimental scans aligned to the reference scans, without altering the position of the experimental scans.

Table 2

Trueness and precision discrepancies measured among the groups tested.

Group	Trueness $\pm$ precision (Overall mean $\pm$ SD maxillomandibular discrepancies) (mm)
Trios4	0.596 ^{a,b} $\pm$ 0.05 ^b
iTero	0.651 ^a $\pm$ 0.04 ^a
i700	0.511 ^{b,c} $\pm$ 0.08 ^c
Modjaw-Trios4	0.453 ^c $\pm$ 0.08 ^c
Modjaw-iTero	0.446 ^c $\pm$ 0.08 ^c
Modjaw-i700	0.423 ^c $\pm$ 0.05 ^c

SD, standard deviation.

Means that do not share a letter are significantly different ($P < .05$).

SDs that do not share a letter are significantly different ($P < .05$).

groups tested and were not significantly different from each other ($P > .05$).

Regarding precision, 1-way ANOVA revealed significant precision discrepancies among the groups tested ($df=5$, $MS=0.053114$, $F=32.21$, $P < .001$) (Fig. 6C). Tukey pairwise comparison test showed significant precision discrepancies among the groups tested. The Modjaw-i700, Modjaw-iTero, Modjaw-Trios4, and i700 groups obtained the best precision values among the groups tested and were not significantly different from each other ($P > .05$) (Table 2). The iTero group obtained the worst precision values among the groups tested and were not significantly different from each other ($P > .05$).

4. Discussion

Based on the results obtained in this in vivo study, the maxillomandibular relationship captured at CR position was influenced by the digital recording method tested. However, when recording the maxillomandibular relationship at CR position, the accuracy of the optical jaw tracking system was not influenced by which of the 3 tested IOSs created the intraoral digital scans that were imported into the jaw tracking system. Therefore, only the first null hypothesis was rejected. Based on these findings, the digital technology used can influence the accuracy of the virtual maxillomandibular relationship acquired. It is important to clarify that this study evaluated the accuracy the mandible

position established with the KD and the polyvinyl siloxane anterior jig positioned in the patient's mouth, not the accuracy of the CR position per se. The anterior jig was used to minimize any variability on the mandible position, for assessing the accuracy of the maxillomandibular relationship captured by the jaw tracker and IOSs tested.

The IOSs tested obtained significant trueness and precision values discrepancies on the maxillomandibular relationship captured at CR position. The i700 and Trios 4 systems obtained the best trueness values, while the iTero had the best precision values. However, the trueness values discrepancies among the IOS groups tested was 140 μ m and the precision values discrepancies among these IOS groups was 40 μ m. The clinical impact of these discrepancies is unknown. Additional clinical studies are needed to further evaluate the impact of the maxillomandibular discrepancies measured at CR position on virtual patient integrations and manufacturing workflow of dental restorations.

An important finding of this clinical study is that the maxillomandibular relationship captured at CR position recorded using the jaw tracking system tested was not impacted by the origin of the intraoral digital scans imported into its program. The maxillomandibular relationship recorded at CR position using the optical jaw tracking system tested obtained higher trueness values compared with the Trios4 and the iTero groups, and similar values to the i700 group. Therefore, the optical jaw tracking system improved the trueness values of the maxillomandibular relationship captured at CR position (by using a Kois deprogrammer and a polyvinyl siloxane anterior jig) compared with the corresponding IOS used to obtain the imported intraoral digital scans. The trueness values discrepancies measured between the IOS and Modjaw groups tested ranged from 143 to 88 μ m. Clinical studies are needed to evaluate the impact of these discrepancies on virtual patient integrations and manufacturing workflow of dental restorations.

In the best knowledge of the authors, there is no previous in vitro or clinical study that assessed the accuracy of the maxillomandibular relationship captured at CR position by using IOSs and an optical jaw tracking system or assessed the influence of the IOS used to obtain the imported scans for the optical jaw tracking system. Therefore, comparisons with previous published data are not feasible.

The KD is one of the described techniques for capturing the CR position [8–12]. The KD technique combined with the polyvinyl siloxane

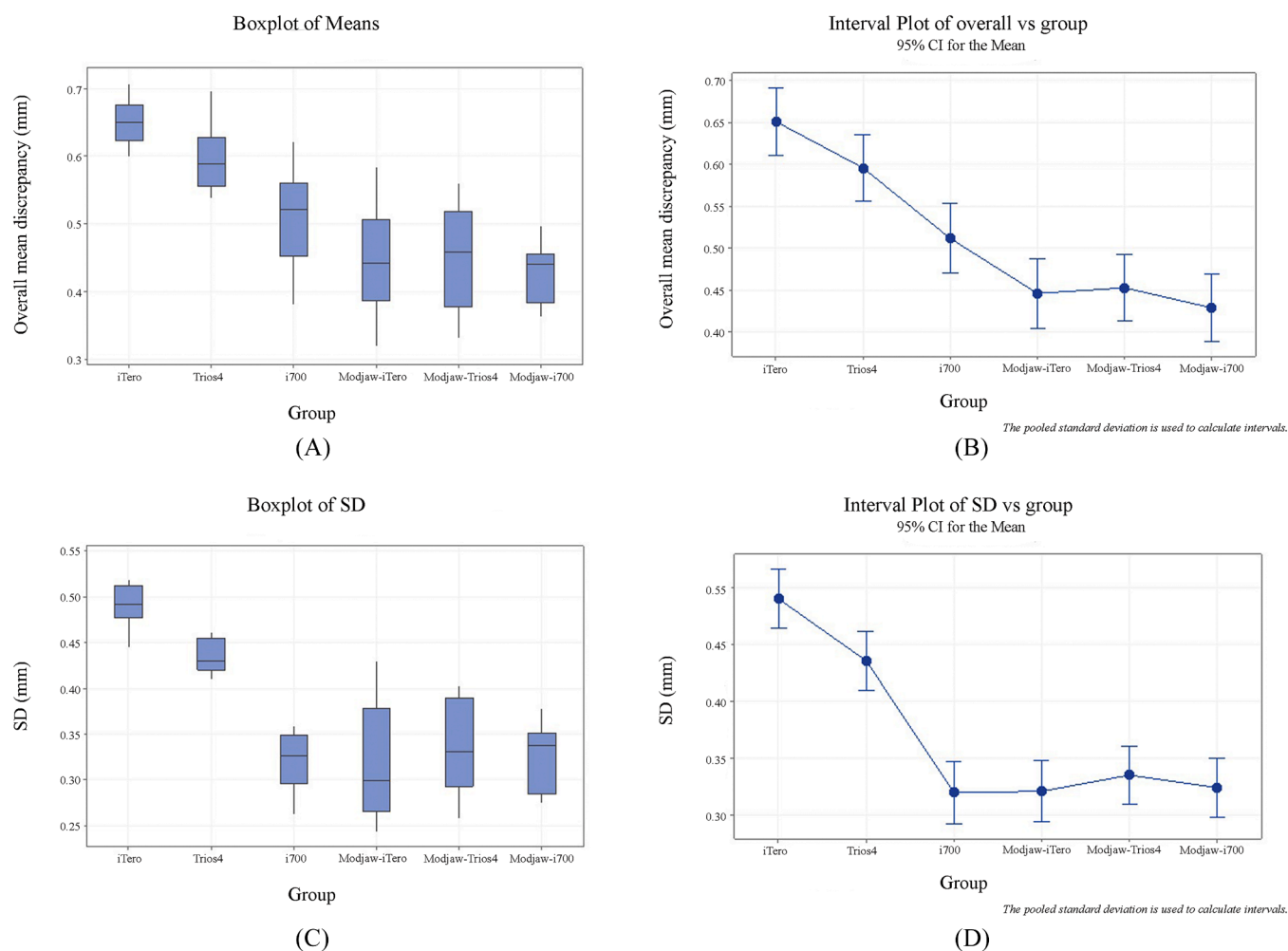


Fig. 6. A, Boxplot of the overall maxillomandibular relationship discrepancies at CR position measured among the groups tested. B, Interval plot of overall maxillomandibular relationship discrepancies at CR position vs group. C, Boxplot of the SDs. D, Interval plot of SDs vs group CR. Centric relation; SD, standard deviation.

anterior jig was selected to eliminate operator manipulation of the mandible [4,12], facilitate a reproducible position of the mandible without tooth contact [1,3], and reduce mandible position changes due to alterations in the body position [6,7]. To reduce mandible position changes due to the participant's fatigue, a KD and the polyvinyl siloxane anterior jig were used, and the data was acquired for only 1 group per day and at the same time on consecutive days. However, psychological variables that can modify muscle tonus, stress, or proprioception could have impacted the results obtained.

Dental literature has evaluated the influence of the virtual records characteristics on the maxillomandibular relationship captured at MIP position by using IOSs [15–18,20,39–41]. Two recent systematic reviews concluded that IOS systems can obtain a similar maxillomandibular relationship at MIP as conventional impression methods in the presence of completely dentate arches, stable occlusal contacts, a single prepared tooth, or arches involving a single missing posterior tooth [39,40]. Additionally, the maxillomandibular relationship at MIP obtained by using IOSs can be influenced by the characteristics of the virtual occlusal records (number, dimension, and location), length of the digitized arches, presence of edentulous areas (position and size), and IOS technology [15–18,20,39–41]. However, in the best knowledge of the authors, there is no previous study that accuracy of the maxillomandibular relationship at CR position recorded by using IOSs. Studies are needed to further evaluate the accuracy of maxillomandibular relationship at CR position recorded by using digital techniques.

A previous *in vitro* study revealed that the bilateral occlusal records

obtained by using an IOS involving 4 teeth obtained higher accuracy on the maxillomandibular relationship at MIP position than bilateral occlusal records involving 2 or 3 teeth [15,20]. In the present clinical study, the bilateral occlusal records recorded by using IOSs included at least 4 teeth each [20]. Additionally, this study attempted to reduce the influence of variables that can reduce intraoral scanning accuracy such as calibrating the IOS before acquiring data [21], controlling the intensity of the ambient light at the patient's mouth [24,25], following the scanning pattern recommended by the manufacturer of the IOS used [26,27], including a completely dentate participant [13,14] without existing restorations [33], controlling the intraoral humidity when obtaining the intraoral digital scans [31], and selecting an experienced operator for capturing the intraoral scanning data [30].

This *in vivo* study has several limitations including the single completely dentate participant tested, limited intraoral clinical conditions tested, and unique CR recording technique utilized. The measurement method used to calculate the accuracy of the maxillomandibular relationship acquired by the jaw tracker and IOSs tested might have introduced some distortion due to the alignment (best fit algorithm) procedure required. Furthermore, analogic procedures may incorporate distortion due to bite registration material shrinkage or mounting stone expansion. Analogic procedures are currently considered the gold standard and used as control group; however, it is unclear if digital procedures may provide a more accurate procedure. Moreover, the validity of the conventional procedures is still unclear. Additional studies are needed to further evaluate the accuracy of these digital

technologies on capturing the maxillomandibular relationship at CR position under varying intraoral conditions such as partially dentate or completely edentulous patients using different CR recording methods.

5. Conclusions

Based on the results and within the limitations (single participant, limited intraoral conditions tested, and unique CR recording technique utilized) of the present in vivo study, following conclusions were drawn:

- The accuracy of the maxillomandibular relationship recorded at CR position was influenced by the choice of digital technique tested.
- Except for the i700 IOS system, the optical jaw tracking system tested improved the trueness value of the maxillomandibular relationship recorded at CR position when compared with the corresponding IOS.
- The IOS used to obtain the intraoral digital scans imported into the optical jaw tracking system tested did not influence the accuracy of the recorded maxillomandibular relationship captured at CR position.
- The Modjaw and the i700 groups obtained the best maxillomandibular relationship trueness values recorded at CR position among the groups tested, and the iTero group had the best precision values among the groups tested.

Clinical implications

The selection among the IOSs and the optical jaw tracking system tested influences the accuracy of the maxillomandibular relationship recorded at CR position using a kois deprogrammer, which might compromise the outcome of prosthodontic treatments.

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Credit author statement

- Marta Revilla-León: Conceptualization, protocol development, data collection, results interpretation, and manuscript writing.
- Lucía Fernández-Estevan: Contributed to the protocol development, IRB submission, and revision of the manuscript.
- Basir A. Barmak: Statistical analysis, data interpretation, and contributed to the manuscript revision.
- John C. Kois: Contributed to results interpretation and manuscript revision.
- Jorge Alonso Pérez-Barquero: Contributed to protocol development, performed measurements, data interpretation, and contributed to the manuscript revision.

All authors discussed the evolution and commented on the manuscript at all stages. All authors approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

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